

inMediaVeritas: A Decentralized News Verification DApp Using zk-SNARKs and Blockchain Consensus

inMediaVeritas Group Project

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Abstract

The proliferation of misinformation in digital media demands trustworthy, censorship-resistant verification mechanisms that operate without relying on any central authority. This paper presents *inMediaVeritas*, a blockchain-based decentralized application (DApp) designed to restore public trust through a three-tier architecture involving verified reporters, community witnesses, and news agencies. The system leverages zk-SNARKs for privacy-preserving location verification, IPFS for decentralized media storage, and on-chain consensus to collectively establish event truthfulness. Beyond the core verification pipeline, we describe three companion applications developed as part of the system: an identity wallet for reporters and witnesses, a simplified authority application for credential issuers, and a news agency subscription interface. We detail the system architecture, cryptographic mechanisms, incentive design, implementation experience, and key limitations.

Keywords: blockchain, zk-SNARKs, decentralized verification, misinformation, IPFS, Ethereum, digital identity, news provenance

1. Introduction

The rapid and largely unchecked spread of false information across digital platforms poses a systemic threat to democratic institutions, public health discourse, and social cohesion [1]. Traditional fact-checking pipelines depend on centralized intermediaries—editorial boards, platform algorithms, or designated verification services which introduce single points of failure, susceptibility to political or commercial bias, and a fundamental asymmetry of trust: users must accept the authority of the verifier without independent recourse.

Decentralized technologies offer a meaningful alternative. By distributing the verification burden across a network of economically incentivized participants whose behavior is governed by cryptographically enforced smart contracts, it becomes possible to construct a verification system that no single party can capture, censor, or manipulate. The blockchain’s immutability provides a permanent, auditable record of every submission, vote, and consensus outcome, while zero-knowledge proofs allow participants to assert facts about themselves (such as their physical proximity to an event) without exposing private data.

inMediaVeritas applies these ideas to the full news lifecycle. Reporters upload media with cryptographic proof of their location at the time of capture. Community members geographically proximate to the event submit testimonials verified through zk-SNARKs. News agencies then pay subscription fees to access and license verified, blockchain-attested media. The result is a pipeline from raw media capture to verified, licensable content that requires no trusted intermediary to arbitrate truth.

This paper makes the following contributions:

1. A complete three-tier architecture for decentralized news verification combining zk-SNARKs, IPFS, and Ethereum smart contracts.

2. A credibility-weighted witness consensus protocol with slashing incentives for dishonest behavior.
3. Three companion applications: an identity wallet for participants, a credential authority application for identity issuers, and a news agency subscription and access interface.
4. A discussion of the practical challenges encountered during implementation and the limitations of the current design.

2. Related Work

2.1 Automated Fact-Checking

NLP-based systems such as ClaimBuster [2] automate claim detection and routing to fact-checkers, but are limited by language ambiguity, domain specificity, and the latency of human review cycles. More recent large-language-model approaches improve fluency and coverage but remain susceptible to confident hallucination and do not produce verifiable provenance records.

2.2 Blockchain Provenance

Blockchain-based provenance systems [3] provide immutable audit trails for digital media, establishing that a given artifact existed at a given time. However, these systems lack community-driven verification of the *events* depicted in the media: provenance confirms what was recorded, not whether it is truthful.

2.3 Zero-Knowledge Proofs

Zk-SNARKs, introduced to the applied cryptography community most visibly through Zcash [4] and subsequently extended by the PLONK proving system [5] or CIRCOM [6] circuits, allow a prover to demonstrate knowledge of a secret witness satisfying a circuit constraint without revealing any information beyond the validity of the statement. This property is critical for protecting the physical locations of journalists and witnesses operating in sensitive or dangerous environments.

2.4 Decentralized Storage

IPFS [7] provides content-addressed, censorship-resistant media storage. Content is identified by its cryptographic hash, meaning any modification to a media artifact produces a different identifier—making tampering immediately detectable.

2.5 Prior Journalism Platforms

The Civil platform [8] combined IPFS with blockchain-based governance for journalism, but lacked a robust community verification mechanism and ultimately faced challenges with adoption and token economics. *inMediaVeritas* addresses both gaps by integrating community witness voting with cryptographically enforced incentives.

3. System Architecture

inMediaVeritas is organized around three interacting actor tiers. All persistent state is anchored to a public blockchain (Ethereum or a compatible L2 network), with media stored on IPFS and referenced by on-chain content identifier (CID) hashes.

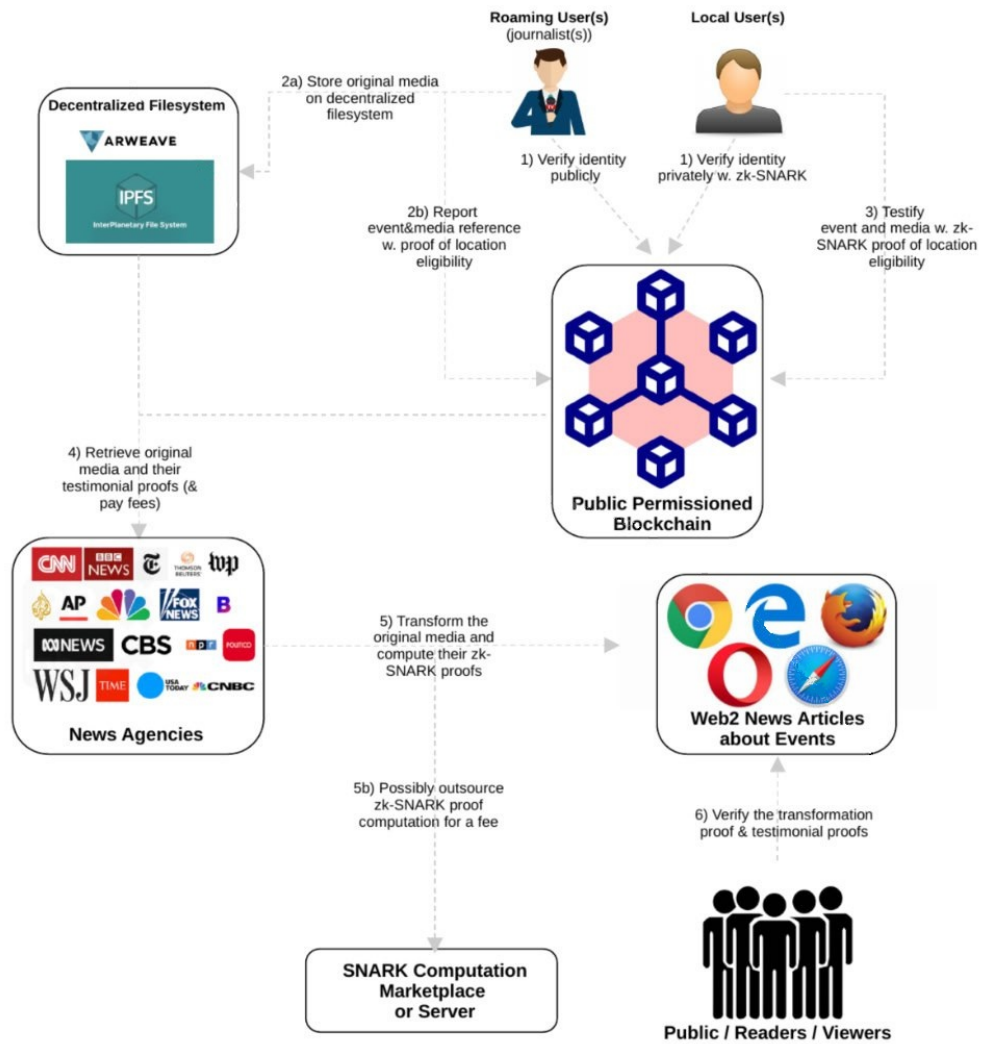


Figure 1: Architecture overview.

3.1 Reporter Layer

Reporters are credentialed participants whose real-world identities are verified through a decentralized identity (DID) registry prior to their first submission. Each news submission bundles three elements committed atomically on-chain via a smart contract: a media artifact stored on IPFS and referenced by its CID, a precise timestamp, and a SHA-256 integrity hash of the raw media. Reporters stake reputation tokens at submission time; demonstrated misconduct triggers slashing, creating a direct economic disincentive for fabricated or misleading reports.

3.2 Witness Layer

Community witnesses are users who are geographically proximate to a reported event. Each witness submits a testimonial (confirm or deny) accompanied by a zk-SNARK proof that their location at the time of the event falls within the claimed event radius, without revealing their precise coordinates. Witness votes are weighted by each witness's historical *credibility score*, which is derived from the alignment of their past votes with the eventual consensus outcomes of resolved events. This mechanism rewards accurate and honest engagement over time, and discourages both coordinated brigading and indiscriminate voting.

The witness contract enforces a minimum quorum before any event is resolved. If quorum is not reached within a configurable time window, the event is marked as unresolved and the reporter's stake is returned. Resolved events are permanently recorded on-chain with the aggregate credibility-weighted vote outcome.

3.3 News Agency Layer

News agencies access verified media and the associated on-chain attestation records by purchasing weekly, monthly, or yearly subscription plans. Subscription fees flow into a shared protocol pool that funds reporter submissions and witness votes, ensuring the system is economically self-sustaining without external grants or centralized governance. Agencies also receive cryptographic proofs, generated on demand, confirming that any published artifact is unaltered and that the associated on-chain record is authentic.

The news agency interface provides subscribing organizations with a dashboard for discovering, filtering, and licensing verified content. Agencies can search the on-chain event registry by location, time range, topic tags, and consensus confidence score. For each event, the interface displays the associated media, the full witness vote breakdown, and a downloadable cryptographic proof of authenticity suitable for inclusion in published reporting. Subscription management, including plan selection and renewal, is handled through a straightforward web front-end that interacts with the subscription smart contract.

4. Companion Applications

In addition to the core smart contract infrastructure, three companion applications were developed to make the system accessible to non-technical participants and to complete the end-to-end user flow.

4.1 Identity Wallet

The identity wallet is a mobile-first application through which reporters and witnesses manage their decentralized identities and interact with the *inMediaVeritas* protocol. It is inspired by EUDI/EIDAS standard for identity wallets in the EU. No suitable existing solution was found, thus it was decided to create a custom tailored application that suits needs of this project. Its primary functions are:

- **Credential storage:** Holding verifiable credentials issued by the authority application (described below), which attest to the user's registration status as a reporter or witness.
- **Submission and voting:** Allowing reporters to compose and submit news events capturing media, attaching a location commitment, and signing the transaction and allowing witnesses to cast credibility-weighted votes with automatically generated zk-SNARK location proofs.
- **Biometric-Protected Signing:** Rather than requiring users to manually manage 256-bit private keys,

the app utilizes *react-native-biometrics*. This allows users to sign protocol messages (reports, votes, and registrations) using keys stored in the device’s Secure Enclave or Keystore, triggered by FaceID or Fingerprint.

4.2 Authority Application

The authority application is a simplified web interface for credential issuers (such as governmental bodies or trusted news organizations) who are responsible for verifying the real-world identities of reporters and witnesses before admitting them to the protocol. Its functions include:

- **Identity verification:** Reviewing submitted government-issued credential proofs and approving or rejecting applicants.
- **Credential issuance:** Issuing signed verifiable credentials to approved participants, which are then stored in their identity wallets and presented when required by the smart contract.
- **Audit log:** Maintaining a tamper-evident log of all issuance actions.

The authority application is deliberately minimal by design: its role is to bootstrap trust at the identity layer without becoming a bottleneck or a central point of control over content decisions. In the future multiple authority applications can be used, (each government can have its own solution) but they all have to support API for communication with the ID app.

5. Cryptographic Mechanisms

5.1 Zero-Knowledge Location Proofs

The core privacy primitive is a zk-SNARK circuit that accepts a witness’s location and their DID-linked identity commitment as private inputs, and proves that (i) the location falls within the declared event radius, and (ii) the identity commitment corresponds to a registered participant, without revealing either the exact location or the underlying identity. We implement this circuit using the CIRCOM proving system [6] over the Baby JubJub elliptic curve [9]—a curve natively optimized for efficient arithmetic inside zk-SNARK circuits and for Ethereum verification. Proof generation runs in a few seconds on consumer hardware; on-chain verification is performed by a Solidity verifier contract generated from the circuit’s verification key.

5.2 Media Integrity

All uploaded media is SHA-256 hashed client-side before upload and the hash is committed on-chain at submission time, creating an immutable fingerprint of the original content. The IPFS CID independently encodes the content hash of the stored artifact. Together, these two independent integrity checks ensure that any tampering with the media after submission is immediately detectable by any party with access to the original artifact.

5.3 Verifiable Credentials and Identity

Reporter and witness identities are anchored to Identity Credentials, each linked to a government-issued credential via a zero-knowledge credential proof. This proof demonstrates that the holder possesses a valid government-issued identity document without revealing the document’s contents or the holder’s name or address. Credentials are issued by registered authority nodes and stored in participants’ identity wallets. The smart contract verifies credential validity on-chain before accepting submissions or votes.

6. Incentive Model

inMediaVeritas is designed to be economically self-sustaining. News agency subscription fees flow into a shared protocol pool, from which: (i) reporters receive a gas reimbursement and a small reward upon successful event resolution, (ii) witnesses receive a proportional share of the resolution reward based on their credibility-weighted contribution to the correct consensus, and (iii) stakes are slashed from dishonest

participants and redistributed to the pool.

This design creates a virtuous cycle: agency demand for verified content directly funds the participation of reporters and witnesses, who are in turn incentivized to report accurately and vote honestly. No external funding source, governance token, or centralized treasury is required to sustain the system.

A key design choice is that reporters and witnesses are not required to provide their own capital. The protocol pool pre-funds participation costs, lowering the barrier to entry for individual contributors. Slashing applies only to pre-funded stakes, not to personal funds, preventing punitive outcomes for good-faith participants who encounter edge cases.

7. Implementation

The system was implemented as a proof-of-concept targeting the Ethereum Sepolia testnet. Smart contracts were written in Solidity and deployed using Hardhat. The zk-SNARK circuits were implemented using the Circom language and compiled with the SnarkJS toolchain, which also generated the Solidity verifier contract. IPFS storage was integrated via the Helia library.

The identity wallet was built as a React Native application, deployment on Android smartphones. The authority application was implemented as a standard React web application. Both front-ends interact with the blockchain via ethers.js and with the IPFS node via the Helia HTTP API.

8. Evaluation

We evaluated the key performance characteristics of the *inMediaVeritas* proof-of-concept. On-chain gas costs were measured on a local EVM (the Remix JavaScript VM), which executes identical bytecode and charges identical gas to a public network such as Sepolia; the measured values are therefore representative of mainnet/testnet execution. zk-SNARK proof generation was measured using the SnarkJS CLI.

8.1 Experimental Setup

On-chain gas measurements were performed on the Remix VM (Solidity 0.8.13, optimizer enabled, `runs=200`, `viaIR=true`). zk-SNARK proof generation was measured with SnarkJS v0.7.6 (Node.js 20.17.0) using the Groth16 proving system on the following platform:

- **Desktop:** AMD Ryzen 5 7600X (6 cores / 12 threads, 4.7 GHz), 32 GB RAM, Windows 11 with WSL 2 (Ubuntu).

The proving curve is BN254 (alt_bn128); the Baby JubJub curve is used only as the embedded curve inside the circuits.

8.2 zk-SNARK Proof Generation

The identity circuit is compiled with the Groth16 proving scheme and contains approximately 7 680 arithmetic constraints. Table 1 reports witness generation and proof generation times. The final Groth16 proof is 192 bytes and is submitted on-chain as part of the `voteOnMediaWithProof` transaction.

Table 1: zk-SNARK proof generation time (Groth16, cold run, SnarkJS v0.7.6).

Circuit	Witness gen. (s)	Proof gen. (s)	Total (s)
Identity	1.121	1.454	2.575
Location	0.951	1.411	2.362

The reported times correspond to a cold run, in which the dominant component is Node.js startup (approximately 0.9 s); the cryptographic work itself accounts for under 0.5 s. In a production setting with a long-running Node process this overhead is amortised, reducing proof generation to an estimated 0.3–0.5 s

per proof. Repeated proving within the same session likewise benefits from the operating system file cache. The two circuits exhibit similar costs, consistent with their comparable constraint counts; the location circuit is marginally faster despite producing a larger public input vector. The measurements were obtained on a high-performance desktop CPU; on weaker laptop-class hardware the same circuits are expected to take several seconds per cold proof, which remains acceptable for an interactive client-side proving workflow.

Circuit setup (Table 2) is performed only once per circuit; every subsequent proof requires only witness calculation and proving.

Table 2: One-time circuit setup cost (performed once per circuit).

Phase	Identity (s)	Location (s)
Circom compilation (r1cs + wasm)	1.872	1.811
Groth16 setup (zkey)	4.877	4.419
zkey contribution	1.675	1.704
Total setup	8.424	7.934

8.3 On-Chain Gas Costs

Table 3 reports the gas consumed by each state-changing protocol operation. At a reference gas price of 10 gwei the corresponding ETH cost per call is also shown. View functions (e.g. `getEvent`, `getMediaItem`) are omitted, as they incur no gas when called off-chain.

Three effects are visible in the data. First, the cost of `createEvent` scales approximately linearly with the number of attached media items ($\approx 150\text{--}170\text{k}$ gas per item), reflecting the on-chain storage of media metadata strings. Second, voting cost depends on storage warmth: the first vote on a given event initialises its counters from zero (cold $0 \rightarrow 1$ storage writes, $\approx 20\text{k}$ gas each), whereas a subsequent vote on the same event updates already-nonzero counters at a reduced cost, lowering the per-vote cost from $\approx 95\text{k}$ to $\approx 61\text{k}$ gas. Third, the `isLocalVoter` flag adds the cost of one additional counter update: when set, the vote increments a separate local-tally counter, and initialising that counter from zero adds a further $\approx 20\text{k}$ gas. This explains why the two `voteOnMedia` measurements differ in the opposite direction to `voteOnEvent`: the first `voteOnMedia` call was cast with `isLocalVoter` disabled (76 190 gas), while the second enabled it, incurring the additional cold local-counter write (81 307 gas).

Table 3: Gas consumption per protocol operation (Remix VM, Solidity 0.8.13, optimizer on).

Operation	Gas (units)	ETH at 10 gwei
Contract deployment	1 649 455	0.016495
<code>createEvent</code> (0 media)	280 774	0.002808
<code>createEvent</code> (1 media)	450 615	0.004506
<code>createEvent</code> (3 media)	735 659	0.007357
<code>addMediaItem</code>	212 567	0.002126
<code>voteOnEvent</code> (first, local)	95 533	0.000955
<code>voteOnEvent</code> (subsequent, local)	61 333	0.000613
<code>voteOnMedia</code> (first, non-local)	76 190	0.000762
<code>voteOnMedia</code> (subsequent, local)	81 307	0.000813

8.4 zk-Enabled Voting Cost

The figures in Table 3 correspond to the non-zk (demonstration) voting mode, in which the voter hash and locality flag are supplied directly. In the zk-enabled mode, a vote is submitted via `voteOnMediaWithProof`,

whose cost is the sum of the storage-update cost measured above ($\approx 76k$ gas) and the cost of on-chain Groth16 proof verification performed by the verifier contract.

The verification cost was measured by deploying the SnarkJS-generated Solidity verifier (via `snarkjs zkkey export solidityverifier`) for each circuit and submitting a real proof produced by the proving pipeline. Both proofs verified successfully on-chain. Table 4 reports the measured verification cost; the figures correspond to the EVM execution cost, which is the relevant quantity when `verifyProof` is invoked internally by `voteOnMediaWithProof` rather than as a standalone transaction.

Table 4: On-chain Groth16 proof verification cost (Remix VM, BN254 curve).

Circuit	Public inputs	<code>verifyProof</code> gas
Identity	5	214 787
Location	7	227 978

As expected for Groth16 on the BN254 curve, verification is dominated by the three elliptic-curve pairing operations and is largely independent of the circuit’s internal size; the small difference between the two circuits ($\approx 13k$ gas) reflects the two additional public inputs of the location circuit, each of which adds one scalar multiplication to the linear combination of the verification key ($\approx 6.5k$ gas per input).

Combining these results, the end-to-end cost of a zk-enabled media vote is approximately $214\,787 + 76\,190 \approx 291\,000$ gas for the identity circuit (and $227\,978 + 76\,190 \approx 304\,000$ gas for the location circuit), of which proof verification accounts for roughly 74–75%. On-chain proof verification, rather than state storage, is therefore the dominant cost in the privacy-preserving voting path. At a reference gas price of 10 gwei this corresponds to approximately 0.0029 ETH per zk-enabled vote.

9. Discussion and Limitations

inMediaVeritas demonstrates that a fully decentralized, privacy-preserving news verification pipeline is technically feasible. However, several limitations merit acknowledgment.

Sparse witness coverage. The system’s correctness depends on a sufficient number of geographically proximate witnesses being present and willing to vote. In low-density areas, rural regions, or for events occurring at unusual times, quorum may be difficult to reach. Mechanisms for incentivizing proactive witness registration in underserved areas remain an open problem.

Authority centralization risk. While multiple authorities can operate in parallel, the credential issuance step reintroduces a degree of centralization at the identity layer. A malicious or captured authority could exclude legitimate reporters or admit Sybil accounts. Federated authority governance and cross-authority credential recognition are mitigations under consideration.

Legal and regulatory uncertainty. The legal landscape for decentralized media licensing is nascent, and smart contract fee enforcement may conflict with jurisdictional copyright, data protection, or media liability frameworks. Production deployment would require careful legal analysis across target jurisdictions.

Scalability. The current implementation targets a single Ethereum L1 or L2 deployment. High-volume news cycles breaking events with thousands of concurrent submissions would stress both the contract gas costs and the IPFS pinning infrastructure. Layer-2 rollups and sharded IPFS pinning clusters are natural directions for scaling.

10. Conclusion

This paper presented *inMediaVeritas*, a decentralized application addressing the misinformation crisis through cryptographic verification, community consensus, and transparent provenance. By combining zk-SNARK

location proofs, IPFS media storage, blockchain immutability, credibility-weighted witness voting, and a carefully designed self-sustaining incentive layer, *inMediaVeritas* enables a trustless pipeline from raw media capture to verified, licensable content without relying on any central authority to arbitrate truth.

The companion identity wallet, authority application, and news agency interface demonstrate that the core protocol can be made accessible to non-technical participants across all three tiers of the system. While significant engineering and sociotechnical challenges remain before production deployment, the proof-of-concept validates the architectural approach and identifies a concrete roadmap for future work.

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